

Last Time:

- Newtonian Review
- Coordinates
- Pendulum

Today:

- State Space
 - Numerical Integration / Simulation
 - Energy
 - Lyapunov Stability
 - Dynamics from Energy
-

State Space

- Pendulum:

$$\ddot{\theta} = -\frac{g}{l} \sin(\theta)$$



- This is a 2nd order ODE. To predict the system's motion we need θ and $\dot{\theta}$ (think initial conditions)
- Let's combine everything we need for prediction into a "state vector":

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}, \quad \underbrace{\dot{x} = f(x)}_{\text{"dynamics"}} = \begin{bmatrix} \dot{\theta} \\ -\frac{g}{l} \sin(\theta) \end{bmatrix}$$

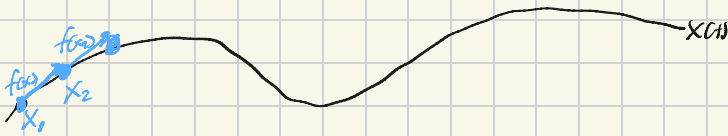
- We can always write ODEs in this 1st-order form
- This is the form that standard ODE solvers use (e.g. ODE45 in Matlab or solve_ivp in Python, etc.)

Numerical Integration / Simulation

- We can't usually solve ODEs analytically $\Rightarrow$ need numerical methods
- Simplest one is Euler integration (aka. "Explicit Euler" or "Forward Euler"):

$$x_{n+1} = x(t_n + \Delta t) \approx x_n + \Delta t \dot{x} = x_n + \Delta t f(x_n)$$

Δt
time step (small)



- Intuition: take small steps along tangent vector ($\dot{x}$) to the solution curve $x(t)$

* Take-Aways from Simulation:

- Simulations can produce unphysical behavior
- Structural properties can change due to discretization error
- You should probably never use explicit Euler
- Always sanity check sim results (e.g. constraints, energy behavior etc.)

Energy

- A scalar quantity that comes in 2 flavours:

* Kinetic ($\sim$ how much I'm moving)

$$T = \frac{1}{2} m \vec{v} \cdot \vec{v} \quad \text{for a particle}$$

* Potential ($\sim$ how much I could move based on where I am)

$$U_{\text{grav}} = m g \underbrace{z}_{\text{height}} \quad \text{for "flat Earth" approximation}$$

- For a particle of $m=1$ that falls $\Delta z=1$ from rest $\Rightarrow$
 $\Delta U = -g \Rightarrow$ gain $T=g$ kinetic

$$\Rightarrow T=g = \frac{1}{2} m \|\vec{v}\|^2 \Rightarrow \|\vec{v}\| = \sqrt{2g}$$

- Other common potentials:

$$U_{\text{spring}} = \frac{1}{2} K x^2 \quad \text{for a linear spring}$$

$\nearrow$ stiffness $\nwarrow$ displacement from equilibrium

$$U_{\text{grav}} = \frac{\overbrace{-G m_1 m_2}^{\text{Gravitational Constant}}}{\underbrace{\|r\|}_{\text{distance between particles}}} \quad \text{for two point masses}$$

* Note: U is only a function of position/configuration (never velocity)

- Total Energy:

$$E = T + U$$

- Forces can be calculated from a potential:

$$\mathbf{F} = -\nabla U(x)$$

$$\Rightarrow F_{\text{spring}} = -\frac{\partial}{\partial x} U_{\text{spring}} = -Kx$$

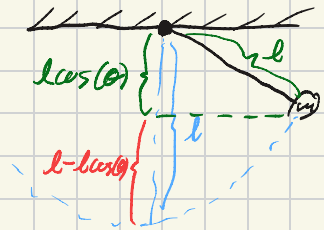
- Forces that can be derived from a potential are called "conservative" since they don't change total energy (just trade for kinetic)
- Friction, drag, and damping are all non-conservative: They depend on velocity and change E .

Lyapunov Stability

- Pendulum Energy:

$$T = \frac{1}{2} m \vec{v} \cdot \vec{v} = \frac{1}{2} m l^2 \dot{\theta}^2$$

$$U = U_{\text{grav}} = m g z = m g l (1 - \cos(\theta))$$



$$\Rightarrow E = T + U = \frac{1}{2} m l^2 \dot{\theta}^2 + m g l (1 - \cos(\theta))$$

* Look $\dot{E}$

$$\dot{T} = m l^2 \dot{\theta} \ddot{\theta} = m l^2 \dot{\theta} \left[\underbrace{-\frac{g}{l} \sin(\theta)}_{\text{plug in dynamics}} \right] = -m g l \sin(\theta) \dot{\theta}$$

$$\dot{U} = \frac{\partial U}{\partial \theta} \dot{\theta} = m g l \sin(\theta) \dot{\theta}$$

$$\Rightarrow \dot{E} = \dot{T} + \dot{U} = 0 \Rightarrow \text{energy is conserved}$$

- Based on this we can bound the system can move in the state space:

* Given E_0 at $t=0$

$$T = \frac{1}{2} m l^2 \dot{\theta}^2 \Rightarrow \|\dot{\theta}(t)\| \leq \sqrt{\frac{2}{m l^2} E_0} \quad \forall t$$

"for all"

$$V = m g l (1 - \cos(\theta)) \Rightarrow \|\theta(t)\| \leq \cos^{-1}\left(1 - \frac{E_0}{m g l}\right) \quad \forall t$$

- When a system is guaranteed to stay within some bounded region we call it "Lyapunov stable"
- We can formalize/generalize this:

* If we can find a scalar-valued function $V(x)$ such that:

$$1) V(x) \geq 0 \quad \forall x, \quad V(0) = 0$$

$$2) V(x) \rightarrow \infty \quad \text{as } \|x\| \rightarrow \infty$$

$$3) \dot{V}(x) = \frac{\partial V}{\partial x} \underbrace{f(x)}_x \leq 0 \quad \forall x$$

$\Rightarrow$ The system is stable around the origin

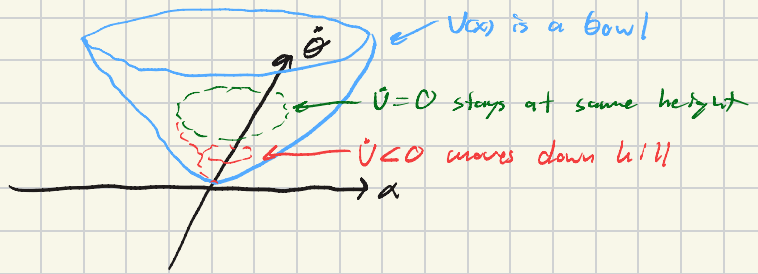
* Note: we can always change coordinates to make some other point x^* the origin:

$$\tilde{x} = x - x^*$$

- $\dot{V}(x) = 0 \Rightarrow$ "Lyapunov stable"

- $\dot{V}(x) < 0 \Rightarrow$ "Asymptotically stable"

* Intuition:



* Pendulum with Euler Integration

- We can do the same Energy/Lyapunov analysis in discrete time looking at $\Delta E_n = E_{n+1} - E_n$ instead of $\dot{E}$:

$$\Delta T_n = T_{n+1} - T_n = \frac{1}{2} m l^2 (\dot{\theta}_{n+1}^2 - \dot{\theta}_n^2)$$

$$\Delta U_n = U_{n+1} - U_n = m g l (\cos(\theta) - \cos(\theta_{n+1}))$$

- Plug in Euler-integrated dynamics:

$$\begin{bmatrix} \theta_{n+1} \\ \dot{\theta}_{n+1} \end{bmatrix} = \begin{bmatrix} \theta_n + \Delta t \dot{\theta}_n \\ \dot{\theta}_n - \Delta t \frac{g}{l} \sin(\theta_n) \end{bmatrix}$$

$$\Rightarrow \Delta E_n = \Delta T_n + \Delta U_n \approx \underbrace{\frac{1}{2} \Delta t^2 m g^2 \sin^2(\theta_n)}_{\text{always positive!}} \geq 0$$

$\Rightarrow$ discretized system is always unstable

Dynamics From Energy

- Let's look at total energy of the pendulum again!

$$E = \frac{1}{2} m l^2 \dot{\theta}^2 + m g l (1 - \cos(\theta))$$

- $\dot{E} = 0$ gives:

$$\dot{E} = m l^2 \dot{\theta} \ddot{\theta} + m g l \sin(\theta) \dot{\theta} = 0$$

- Divide by $m l \dot{\theta}$:

$$\ddot{\theta} + \frac{g}{l} \sin(\theta) = 0$$

same dynamics!

- * We got the dynamics via conservation of energy without having to calculate $\vec{F}$ or $\vec{a}$ via coordinate transforms!
- * Unfortunately this doesn't work in general (e.g. for non-conservative systems and multi-DOF systems).